# 清华大学热能工程系 2010 级“燃烧理论”期中考试试卷（A 卷）

# Middle Term Examination of Combustion Theory (A)

**考试时间(Date)：　　2013 年 4 月 26 日班级(Class)：**

**学号(Student No.)：姓名(Name)：**

**Instructions(提示):**

1　请将你的学生证放在桌子的右上角（Please put your Photo student ID on the desk）

2　考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）

3　试卷共有7题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有6页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are 11 questions on this exam with 5 pages. You have up to 120 minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4　如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5　强烈推荐你先读一下所有的问题，并对回答问题的时间进行安排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6　请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7　答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

**1. 绝热燃烧温度的计算**(18 分)

1) 假设产物中没有离解发生，计算下面公式所示的化学反应在 298K，一个大气压情况下的绝热火焰温度。

$$CH_4 + 13(0.21O_2 + 0.79N_2) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 + n_{O_2}O_2$$

2) 将反应物中的氮气 $N_2$ 替换为 $CO_2$，并再次计算绝热火焰温度，并简要分析二者不同的原因。

1.The prediction of adiabatic combustion temperature (18 points)

1） Assuming no dissociation of product species, determine the adiabatic flame temperature for the following chemical reaction at $T_i = 298k$ , $P$=1atm

$$CH_4 + 13(0.21O_2 + 0.79N_2) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 + n_{O_2}O_2$$

2） Change the $N_2$ to $CO_2$. Determine the adiabatic flame temperature again， and briefly explain the difference between them.

解答：1） $CH_4 + 13(0.21O_2 + 0.79N_2) \rightarrow CO_2 + 2H_2O + 10.27N_2 + 0.73O_2$ 2

$$H_{react}(T_i,\ P) = H_{prod}(T_{ad}, P)$$

Assume Cp @1200K

$$H_{react} = -74831kj$$

$$\begin{aligned} H_{pro} &= (-393546 + (T_{ad} - 298) \times 56.21) + 2(-241845 + (T_{ad} - 298) \times 43.87) + \\ &= 10.27((T_{ad} - 298) \times 33.71) + 0.73((T_{ad} - 298) \times 35.68) \qquad 3 \\ &= -877236 + (T_{ad} - 298) \times 516.1981 \end{aligned}$$

$T_{ad} = 1852.45K$ 3

2) $CH_4 + 13(0.21O_2 + 0.79CO_2) \rightarrow 11.27CO_2 + 2H_2O + 0.73O_2$

$$H_{react}(T_i,\ P) = H_{prod}(T_{ad}, P)$$

Assume Cp @900K

$$H_{react} = -74831 - 10.27 \times 393546 = -4116548.42$$

$$\begin{aligned} H_{pro} &= 11.27(-393546 + (T_{ad} - 298) \times 53.136) + 2(-241845 + (T_{ad} - 298) \times 39.93) \\ &\quad + 0.73((T_{ad} - 298) \times 34.454) \\ &= -4918953.42 + (T_{ad} - 298) \times 703.854 \end{aligned}$$

$T_{ad} = 1438K$

The main reason is that the CO2 owns higher heat capacity.

**2.** 有一个由扩散火焰形成的反应器系统，气流速度为 2.5m/s，反应器截面直径为 50mm，反应器温度为 1200K，其中 $D = 2.2\times10^{-5} m^2/s(273K, 1atm)$，$Sc = \frac{\nu}{D} = 0.8$，分别计算其贝克利数( $Pe = \frac{\nu L}{D}$ )、雷诺数，并通过贝克利数判断这个反应器是否是柱塞流反应器 (10 分)。

2. A non premixed flame reactor was a one dimensional flow reactor. The gas velocity is 2.5m/s. The cross section diameter of the reactor is 50 mm. $D = 2.2\times10^{-5} m^2/s(273K, 1atm)$, $Sc = \frac{\nu}{D} = 0.8$. Calculate the *Pe*clet number $Pe = \frac{\nu L}{D}$ and Reynolds number, and judge whether the reactor is a plug flow reactor.(10 points).

解答： $Pe = \frac{2.5\times0.05}{2.2\times10^{-5}\times(1200/273)^{1.5}} = 616.53$ 4

$$\mathrm{Re} = \frac{\rho V L}{u} = \frac{1}{Sc\times D}\times\frac{VL}{1} = \frac{2.5\times0.05}{0.8\times2.2\times10^{-5}\times(1200/273)^{1.5}} = 771$$ 4

贝克利数较大，表明对流强度强，可判断其为柱塞流反应器。 2

**3.**燃料液滴蒸发对实际装置具有重要意义。考虑一个直径 100μm 的正癸烷（$C_{10}H_{22}$）液滴在常温(298K)、1atm 的干燥氮气中蒸发的情形。液滴表面温度为 25℃。试判断采用下式

$$K = \frac{8\bar{\rho}D_{C_{10}H_{22},N_2}}{\rho_l}\frac{Y_{C_{10}H_{22},s} - Y_{C_{10}H_{22},\infty}}{1 - Y_{C_{10}H_{22},s}}$$

来计算蒸发速率常数 K 是否合理（假设此时液滴表面温度恒定）？并说明原因。(12 分)

注：上式中 $\rho_l$ 是液滴密度，$\bar{\rho}$ 是气体混合物平均密度，$D_{C_{10}H_{22},N_2}$ 是正癸烷在氮气中的二元扩散系数，$Y_{C_{10}H_{22},s}$、$Y_{C_{10}H_{22},\infty}$ 分别为液滴表面和无穷远处的正癸烷蒸气质量分数。取 $Y_{C_{10}H_{22},\infty} = 0$。计算可能用到的参数见下表。

| 燃料<br>fuel | MW<br>（kg/kmol） | $\rho_l$ （kg/m 3) | 沸点<br>boiling point（℃） | $h_{fg}$<br>（kJ/kg） |
|---|---|---|---|---|
| 正癸烷 n-Decane | 142.28 | ~730 | 174.1 | 277 |

3. The evaporation of fuel droplet is important for practical devices. Here we consider a droplet of

n-decane ($C_{10}H_{22}$) of a diameter of 100μm in dry nitrogen ambience (298K , 1atm). The temperature of droplet surface is the same with the ambience. Would it be reasonable to calculate the evaporation rate constant *K*by using

$$K=\frac{8\bar{\rho}D_{C_{10}H_{22},N_2}}{\rho_l}\frac{Y_{C_{10}H_{22},s}-Y_{C_{10}H_{22},\infty}}{1-Y_{C_{10}H_{22},s}}$$

The surface temperature remains unchanged during the evaporation.Please explain it (12 points).

$\rho_l$ is the droplet density, while $\bar{\rho}$ is mean density of gas mixtures. $D_{C_{10}H_{22},N_2}$ is the binary diffusion coefficient between n-decane and nitrogen. $Y_{C_{10}H_{22},s}$ and $Y_{C_{10}H_{22},\infty}$ are the mass fractions of n-decane gas near the droplet surface and at infinity, respectively. Assume that $Y_{C_{10}H_{22},\infty}=0$. The possibly used physical properties are listed in the table above.

解答：直接计算传质数 $B_Y$ 即可。需要首先计算液滴表面处的燃料质量分数。利用积分形式的 Clausius-Clapeyron 方程计算：

$$\frac{P_{sat}}{P(=1atm)}=\exp\left[\frac{-277000}{8315/142.284}\left(\frac{1}{298}-\frac{1}{447.1}\right)\right]=4.97\times10^{-3}$$

即 $x_A=4.97\times10^{-3}$，由此计算

$$Y_{A,s}=4.97\times10^{-3}\times\frac{142.284}{4.97\times10^{-3}\times142.284+(1-4.97\times10^{-3})\times28.014}=0.0247$$

$$B_Y=\frac{Y_{A,s}-Y_{A,\infty}}{1-Y_{A,s}}=\frac{0.0247-0}{1-0.0247}=0.0254<<1$$

实际上 $\ln(1+B_Y)=0.02505\approx B_Y$，因此可以认为这样近似是合理的。

如果还要算下去，则需要计算气液平均温度下的二元扩散系数、表面及无穷远处的平均气体密度，代入计算 K 值。

但实际中由于重大疏忽，在题目中给出的公式错误，成了 $K=\frac{8\bar{\rho}D_{C_{10}H_{22},N_2}}{\rho_l}\frac{1-Y_{C_{10}H_{22},\infty}}{1-Y_{C_{10}H_{22},s}}$

故而造成结果与原意完全不同。实际批改中给分如下：

首先（或最后）判断不合理　　3 分；

利用 Clausius-Clapeyron 方程最终计算出 $Y_{A,s}$ 或进一步计算 $B_Y$　　6 分

利用 $B_Y$ 说明 ln(1+BY)和 1+BY 根本不同；或利用 K 的正确公式和题给公式分别计算，

比较二者的巨大差异（40 倍），说明近似完全不合理。　　3 分。

**4.**将氮氧化物生成的热力型（或泽尔多维奇，Zeldovich）机理简化为以下一步反应：

$$N_2 + O_2 \Leftrightarrow 2NO$$

计算该反应在 1600K、2400K 下的标准平衡常数。根据计算结果，比较热力型 $NO_x$ 生成机理与温度相关的变化。**本题要求：只能采用下表中列出的各物质的热力学数值**。(14 分)

4. The thermal (or Zeldovich) mechanism of NOx formation is expressed by a one-step reaction:

$$N_2 + O_2 \Leftrightarrow 2NO$$

Please calculate the standard equilibrium constants of the reaction at 1600K and 2400K, respectively. Discuss the role of thermal mechanism during NOx formation according to the results.

**Attention: You can ONLY use the thermodynamical data listed in the table below.**(14 points)

氮气（$N_2$），标准生成焓=0

Nitrogen ($N_2$), standard enthalpy of formation = 0

| T/K | $\left(\bar{h}^0(T) - \bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 41899 | 244.048 |
| 2400 | 70645 | 258.598 |

氧气（$O_2$），标准生成焓=0

Oxygen ($O_2$), standard enthalpy of formation = 0

| T/K | $\left(\bar{h}^0(T) - \bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 44253 | 260.339 |
| 2400 | 74467 | 275.627 |

一氧化氮(NO)，标准生成焓= 90297 kJ/kmol

Nitric oxide (NO), standard enthalpy of formation = 90297 kJ/kmol

| T/K | $\left(\bar{h}^0(T) - \bar{h}_f^0(298)\right)$ /(kJ/kmol) | $\bar{s}^0(T)$ /(kJ/(kmol`K)) |
|---|---|---|
| 1600 | 43317 | 264.937 |
| 2400 | 72617 | 279.772 |

解答：本题比较简单，主要利用公式：$K_p^{\Theta} = \exp\left(-\frac{\Delta G_T^0}{R_u T}\right)$　　2 分

而在本题中，计算 $\Delta G_T^0$ 必须采用：$\Delta G_T^0 = \Delta H^0 - T\Delta S^0$　　2 分

而绝对焓由两部分构成，即 $\bar{h}^0(T) = \bar{h}_f^0(298) + \left(\bar{h}^0(T) - \bar{h}_f^0(298)\right)$

计算结果：（计算过程 2*2 = 4 分，结果 2*2= 4 分）

| T/K | $\Delta\bar{H}^0$ | $\Delta\bar{S}^0$ | $\Delta G_T^0$ | $K_p^0$ |
|---|---|---|---|---|
| 1600 | 181076 | 25.487 | 140296.8 | 2.63e-5 |
| 2400 | 180716 | 25.319 | 119950.4 | 2.45e-3 |

可见该反应的标准平衡常数随温度下降而迅速减小，故而从热力学角度说明，热力型 NOx 生成机理只在高温下重要。　　2 分

**5. 研究下列一组反应：**(15 分)

$M+Br_2 \rightarrow 2Br+M$　(1)

$Br+H_2 \rightarrow HBr+H$　(2)

$H+Br_2 \rightarrow HBr+Br$　(3)

$H+HBr \rightarrow H_2+Br$　(4)

$Br+Br+M \rightarrow Br_2+M$ (5)

1) 试说明上述反应中，哪些是链的激发反应，哪些是链的传递反应，哪些是链的中断反应（3 分）
2) 写出上述反应的总包反应。（2 分）
3) 试写出 d[HBr]/dt，要求用[$H_2$]和[$Br_2$]表示[提示：假设 Br 和 H 处于稳态条件]。（10 分）

解答：

1）激发：1；传递：2,3,4；中断：5

2）$H_2 + Br_2 \rightarrow 2HBr$

3）$d[HBr]/dt = k_2[Br][H_2] + k_3[H][Br_2] - k_4[H][HBr]$　2 分

稳态条件：

$d[Br]/dt = 2k_1[M][Br_2] - k_2[Br][H_2] + k_3[H][Br_2] + k_4[H][HBr] - 2k_5[Br]^2[M] = 0$　2 分

$d[H]/dt = k_2[Br][H_2] - k_3[H][Br_2] - k_4[H][HBr] = 0$　2 分

之后的推导过程　2 分

结果：　$d[HBr]/dt = \dfrac{2k_2k_3[H_2][Br_2]}{k_3[Br_2]+k_4[HBr]}\left\{\dfrac{k_1}{k_5}[Br_2]\right\}^{\frac{1}{2}}$　2 分

5. As for the reactions bellows: (15 points)

$M+Br_2 \rightarrow 2Br+M$　(1)

$Br+H_2 \rightarrow HBr+H$　(2)

$H+Br_2 \rightarrow HBr+Br$　(3)

$H+HBr \rightarrow H_2+Br$　(4)

$Br+Br+M \rightarrow Br_2+M$　(5)

(1) Please point out the chain-initiation, chain-propagating and chain-terminating reactions.
(2) Write the global reaction rate of above reactions
(3) Write the expression of d[HBr]/dt, only in terms of [$H_2$] and [$Br_2$] [Hint: assume that Br and H are in steady state situation].

**6. 化学反应净生成率 $w_i$ 的理解**(16 分)

(1) 某组分 $i$ 净生成率 $w_i$ 单位是 kmol/m$^3$-s, 你认为其定义式 $w_i = \frac{d[N_i / V]}{dt}$ 正确吗？并给出解释（4 分）。

(2) 全混流反应器认为反应器内返混无穷大，即扩散十分快，从而导致反应器内各点和出口的参数性质一致，而组分守恒方程却可写成 $w_i + \frac{\dot{m}}{\rho V}([X_i]_{in} - [X_i]_{out}) = 0$，即反应器进、出口化学反应项和对流项的平衡，反而**并没有含扩散项**，您是如何理解的？(6 分)

(3) 对于反应 O + O + M → $O_2$ + M，其中 M 是第三体，其浓度$[M]$保持不变。假设反应速率常数为 $k$。试确定反应速率 $w_O$ 的表达式，并推导该反应的化学特征时间尺度 $\tau_{chem}$ 的表达式。（均由$[O]$、$[M]$及 $k$ 来表达）(6 分)

6. The understanding of some issues related the net chemical production rate $w_i$. (16 points)

(1) $w_i$, of species i, has a unit of kmol/m$^3$-s, do you think whether the definition of $w_i = \frac{d[N_i / V]}{dt}$ is right or not？Please explain it（4 points）。

(2) The well-stirred reactor has the ideally perfect mixing in the control volume, implying the diffusion is greatly larger than the convection in the reactor. So, the composition of species within the reactor is everywhere the same, while the composition at the outlet must be the same as in the interior. The conservation equation of species $i$ can be expressed as $w_i + \frac{\dot{m}}{\rho V}([X_i]_{in} - [X_i]_{out}) = 0$. Note that, **there is no diffusion term, the larger term, in the above equation**. How can you understand it? And explain it. (6 points)

(3) For elementary reaction of O + O + M → $O_2$ + M，where M is an arbitrary species and $[M]$ remains unchanged.Assuming rate constant $k$, please write the rate expression of $w_O$，and then determine the characteristic time of the reaction, $\tau_{chem}$on the basis of $[M]$, $[O]$ and$k$. (6points)

解答：1）不能。2 分

$$\frac{\mathrm{d}[x_i]}{\mathrm{d}t}=\frac{\mathrm{d}[N_i/V]}{\mathrm{d}t}=\frac{1}{V}\frac{\mathrm{d}N_i}{\mathrm{d}t}-N_i\frac{1}{V^2}\frac{\mathrm{dV}}{\mathrm{d}t}$$

$$\frac{\mathrm{d}[x_i]}{\mathrm{d}t}=\dot{\omega}_i-[x_i]\frac{1}{V}\frac{\mathrm{dV}}{\mathrm{d}t}$$

从上面得公式中可以得出结论。2 分

2) 内部扩散无限大 2 分

外部对流大于扩散 2 分

3） $\frac{d[O]}{dt}=(-2k[m])[O]^2$ 3 分

$\tau=\frac{(e-1)}{[O]2k[M]}$ 3 分

7．甲醇（$CH_3OH$）被认为是一种很有潜力的含氧燃料，其氧化燃烧应该很复杂，显然要全面描述这一系统是很困难的。你是否可以大致描述一下主要的氧化燃烧的过程（假设是在高温条件下，温度为 1800K）。（15 分）

7. Though methanol （$CH_3OH$）is new fuel containing oxygen and its oxidation should be very complicated, which causes it very difficult to completely describe all these reaction mechanisms. Please briefly summarize the oxidation mechanism of methanol.

解答：基本有写对应 5 分

去氢过程写对 3 分 CH2OH CH3O by(H OH)

CH2O - > HCO 5 分

CO - > CO2 2 分